

Response to the Editor

Information Sciences

Manuscript No. INS-D-26-5725

Fisher–Rao Geometric Optimization and Detection for Generative Steganographic Transport

Stéphane Gaël R. Ekodeck, Prosper Rosaire Mama Assandje, Serge Alain Ébébé,
Stéphane Willy Mossebo Tcheunteu, Arthur Ulrich Ewane, Damaris Belle M. Fotso, René
Ndoundam

Submitted: 2026 | Revised: June 3, 2026

Dear Editor,

We are grateful for the careful engagement with our manuscript and for the opportunity to resubmit a revised version. Below we address every point raised in full, describe precisely where the manuscript has been modified, and include a summary table of all changes.

1. Editorial Concern and Response

1.1 Core editorial concern

Editor’s question (condensed): “The use of Fisher–Rao geometry, natural-gradient methods, and information-geometric optimization frameworks are themselves well-established mathematical tools. What is it that makes it different from the previous work? The manuscript did not demonstrate a sufficiently compelling advancement relative to the current state of the literature and the high novelty threshold maintained by *Information Sciences*.”

Our response. We fully accept this as the right question and have restructured the introduction to answer it directly and explicitly. The tools (Fisher–Rao metric, α -connections, natural gradient) are indeed established. Our contribution is not that these tools exist, but that, when applied to the *steganographic transport problem*, they yield concrete results that are neither obvious nor previously known. We detail four such results below.

2. What the Framework Produces That Is Genuinely New

2.1 A new problem formulation

Gap in the prior literature. Every steganographic method prior to this work — from Cachin’s ε -security criterion to state-of-the-art coverless schemes iMEC and Discop — treats embedding as a *point-to-point* problem: it minimises the endpoint divergence $D_{\text{KL}}(P_{\theta_1} \| P_{\theta_0})$ between the final stego distribution and the cover distribution, while ignoring how the generator parameter evolves from θ_0 to θ_1 .

We are the first to ask: *does the trajectory through parameter space matter for the accumulated detectability of a multi-step embedding process?* This question has no precursor in either the steganography or the information-geometry literature. The answer, formalised by the sequential-KL path-cost functional

$$\mathcal{C}[\gamma] = \sum_{k=0}^{N-1} D_{\text{KL}}(P_{\gamma(t_{k+1})} \parallel P_{\gamma(t_k)}),$$

is yes: the accumulated detectability depends on the entire trajectory, and the optimal path is *not* the Euclidean straight line. The formulation of this path-cost problem is a conceptual contribution independent of the tools used to solve it.

2.2 Geodesic optimality (Theorem 2, part iv): a non-trivial result

Possible confusion: “Geodesics minimise length” is a tautology. Is Theorem 2 anything more?

Yes. The steganographic objective — minimising $\mathcal{C}[\gamma]$ over a discretised path — is *not* the Riemannian length functional. These two quantities coincide only in the limit of infinitesimally fine discretisation and under a specific second-order KL expansion.

Theorem 2(iv) proves the following chain:

1. The local KL expansion gives $D_{\text{KL}}(P_{\theta+\delta} \parallel P_{\theta}) = \frac{1}{2} \delta^{\top} F(\theta) \delta + O(\|\delta\|^3)$.
2. Summing over a discretised path, the leading-order term is the Fisher path-energy $E[\gamma] = \int_0^1 \dot{\gamma}^{\top} F(\gamma) \dot{\gamma} dt$.
3. Minimising $E[\gamma]$ over paths with fixed endpoints yields the geodesic equation as the Euler–Lagrange condition.

This chain links the steganographic path cost to the Riemannian structure through a specific asymptotic argument. It does not appear in the natural-gradient or information-geometry literature (where the path-cost problem for KL accumulation has not been studied) nor in the steganography literature (which does not consider trajectory geometry). A **Remark** to this effect has been added directly before Theorem 2 in the revised manuscript.

2.3 Detection Duality (Theorem 7): a structural result specific to steganography

Possible confusion: The dual-affine structure of exponential families (*e*-geodesics, *m*-projections) is classical (Amari–Nagaoka). Is Theorem 7 anything more?

Yes. The classical result is that exponential families admit a dually flat structure. What is new is the *adversarial assignment* of the two geometries:

- The **embedder**’s optimal path lives in the *Levi-Civita* ($\nabla^{(0)}$, Fisher–Rao) geometry — governed by Theorem 2.
- The **optimal Neyman–Pearson steganalyzer**’s rejection region is affine in expectation coordinates (*m*-flat half-space), i.e. it lives in the *e-geometry* ($\nabla^{(+1)}$)

— governed by Theorem 7.

The Amari–Chentsov ± 1 connection pair thus plays *opposite and complementary roles for the two adversaries*. This structural ± 1 duality is specific to the steganographer–steganalyzer adversarial setting and has no counterpart in prior work on exponential families or on steganographic security. An introductory paragraph making this explicit has been added before Theorem 7 in the revised manuscript.

2.4 Stein exponent (Theorem 8): closing the loop between embedding and detection geometry

Possible confusion: Stein’s lemma identifying $-D_{\text{KL}}(P_{\theta_s} \| P_{\theta_c})$ as the optimal type-II error exponent is classical.

The classical Stein result is well known. What Theorem 8 adds in the steganographic context is the *chain of connections*:

$$\beta_n^{(\alpha)} \lesssim \exp\left(-\frac{n}{2} d_g(\theta_c, \theta_s)^2\right),$$

obtained by expressing the KL divergence through the local Fisher–Rao distance expansion (Theorem 6). This closes the loop between the two adversaries:

- A small Fisher–Rao distance $d_g(\theta_c, \theta_s)$ buys low accumulated embedding distortion (Theorem 2) *and* high type-II error for the steganalyzer (Theorem 8).
- The same geometric quantity governs both the embedding cost and the detection difficulty.

This unification of embedding geometry and detection geometry through a single Fisher–Rao distance is new. It does not appear in the endpoint-only steganographic security literature (Cachin, Fridrich–Ker), which provides no connection to the trajectory. A **Remark** to this effect has been added before Theorem 8 in the revised manuscript.

2.5 The two faces of curvature: a new adversarial trade-off

In the regime of negative sectional curvature (which includes the Gaussian family, $\kappa = -1/2$), curvature plays *opposite roles for the two adversaries simultaneously*:

- **Detection advantage:** negative curvature expands the Fisher-ball volume accessible to the steganalyzer (amplified Jacobi-field growth), increasing the resolution of the detection test.
- **Embedding cost:** the same negative curvature amplifies geodesic transport instability for the embedder.

This geometric trade-off, intrinsic to the adversarial game, is a new observation with no precursor in either the steganography or the Riemannian geometry literature.

3. Summary of All Changes to the Manuscript

No theorems, proofs, simulations, or conclusions have been altered. The scope statement (theoretical foundation, not operational pipeline) is unchanged.

Table 1: Complete list of modifications in the revised manuscript.

Location	Nature of change	Addresses
§1.4 (new subsection)	“Beyond the Tools: What the Framework Produces” — 4 focused paragraphs answering the editor’s central question.	Core editorial concern
Table 1 (new)	10-row $\times$ 5-column positioning table: contributions vs. prior literature (Cachin/Fridrich-Ker; iMEC/Discop/SparSamp; Amari IG; natural-gradient optimisation; this paper). Four exclusive contributions are in bold.	Core editorial concern
Before Theorem 2	Remark explaining why part (iv) is not a tautology of “geodesics minimise length,” with explicit proof sketch of the second-order KL $\rightarrow$ Fisher-energy chain.	Non-triviality of key results
Before Theorem 7	Introductory paragraph clarifying that the novelty is not the classical dual-affine structure but its adversarial steganographic interpretation (± 1 geometry assigned to embedder vs. steganalyzer).	Non-triviality of key results
Before Theorem 8	Remark distinguishing from classical Stein’s lemma and explaining the new chain: Stein exponent $\leftrightarrow$ Fisher-Rao distance $\leftrightarrow$ embedding cost.	Non-triviality of key results

4. Positioning vs. Prior Literature

To make the novelty self-evident, the revised manuscript now includes Table 1 (reproduced below as Table 2). Rows in **bold** represent capabilities exclusive to this paper.

Closing Remarks

We believe the revised manuscript now answers the editorial question directly, explicitly, and with concrete precision at every relevant location in the text. The four contributions exclusive to this paper (bold rows in Table 2) are non-trivial consequences of applying established tools to a genuinely new problem formulation — the steganographic transport path problem — and to a genuinely new adversarial setting in which the ± 1 dual geometry plays structurally distinct roles for the two adversaries.

We hope the revised manuscript meets the novelty threshold expected by *Information Sciences* and look forward to your assessment.

Respectfully yours,

Stéphane Gaël R. Ekodeck
Corresponding author

Table 2: Positioning of contributions relative to prior literature ($\checkmark$ = provided; $\circ$ = partially; $\times$ = not provided).

Capability	Cachin / F-K	iMEC / Discop	Amari IG	Nat. grad.	This paper
KL endpoint security criterion	$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$
Coverless / generative embedding	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$
Fisher–Rao metric on param. space	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
Natural-gradient optimisation	$\times$	$\times$	$\circ$	$\checkmark$	$\checkmark$
Path-cost formulation	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Sequential-KL geodesic optimality (Thm. 2)	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Detection-duality (± 1 structure, Thm. 7)	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Curvature trade-off in adversarial game	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Convergence guarantees for stego optimizer	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
Numerical illustrations on stat. manifolds	$\times$	$\circ$	$\times$	$\circ$	$\checkmark$

University of Yaoundé I — LIRIMA/GRIMCAPE & UMMISCO, Sorbonne Universités
 stephane-gael.ekodeck@facsciences-uy1.cm

On behalf of all co-authors:

P.R. Mama Assandje, S.A. Ébélé, S.W. Mossebo Tcheunteu, A.U. Ewane, D.B.M. Fotso, R. Ndoundam.